



# LESSON - 4





# SIKHOGE KYA???

## INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

## DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

## SQL STATEMENT FUNDAMENTALS

- SELECT \* FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



## DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

## JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

## GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

## ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES



| Data Type     | Description                                                            | Usage Example          |
|---------------|------------------------------------------------------------------------|------------------------|
| INT           | Integer (whole numbers)                                                | age INT                |
| TINYINT       | Small integer, usually between -128 and 127                            | rating TINYINT         |
| SMALLINT      | Small integer, usually between -32,768 and 32,767                      | score SMALLINT         |
| BIGINT        | Large integer, for values beyond INT range                             | population BIGINT      |
| DECIMAL(p, s) | Fixed-point number with p precision and s scale (digits after decimal) | price DECIMAL(10, 2)   |
| NUMERIC(p, s) | Same as DECIMAL, for fixed-point numbers                               | balance NUMERIC(12, 4) |
| FLOAT         | Floating-point number with single precision                            | average_score FLOAT    |
| DOUBLE        | Double precision floating-point number                                 | distance DOUBLE        |
| BIT           | Boolean type (1 for true, 0 for false)                                 | is_active BIT          |
| VARCHAR(n)    | Variable-length string (up to n characters)                            | name VARCHAR(100)      |
| CHAR(n)       | Fixed-length string (padded with spaces)                               | code CHAR(5)           |
| TEXT          | Large text string (variable length)                                    | description TEXT       |
| LONGTEXT      | Larger text data than TEXT                                             | comments LONGTEXT      |
| BLOB          | Binary Large Object for storing images, files, etc.                    | image BLOB             |

# Data Definition Language (DDL)

"DDL sets up your database structure—tables, indexes, and schemas. It’s how you create, modify, or delete the blueprint of your database!"

| Command  | Description                                                                                   | Syntax                                                               |
|----------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| CREATE   | Create database or its objects (table, index, function, views, store procedure, and triggers) | CREATE TABLE table_name (column1 data_type, column2 data_type, ...); |
| DROP     | Delete objects from the database                                                              | DROP TABLE table_name;                                               |
| ALTER    | Alter the structure of the database                                                           | ALTER TABLE table_name ADD COLUMN column_name data_type;             |
| TRUNCATE | Remove all records from a table, including all spaces allocated for the records are removed   | TRUNCATE TABLE table_name;                                           |
| COMMENT  | Add comments to the data dictionary                                                           | COMMENT 'comment_text' ON TABLE table_name;                          |
| RENAME   | Rename an object existing in the database                                                     | RENAME TABLE old_table_name TO new_table_name;                       |

| Data Type    | Description                                                          | Usage Example                          |
|--------------|----------------------------------------------------------------------|----------------------------------------|
| DATE         | Date in YYYY-MM-DD format                                            | birthdate DATE                         |
| TIME         | Time in HH:MM:SS format                                              | start_time TIME                        |
| DATETIME     | Date and time in YYYY-MM-DD HH:MM:SS format                          | created_at DATETIME                    |
| TIMESTAMP    | Automatically tracks date and time of record creation/update         | last_updated TIMESTAMP                 |
| YEAR         | Stores a 4-digit year                                                | join_year YEAR                         |
| BOOLEAN      | Stores TRUE or FALSE values (often mapped to TINYINT(1))             | is_active BOOLEAN                      |
| BINARY(n)    | Fixed-length binary data (n bytes)                                   | image_data BINARY(256)                 |
| VARBINARY(n) | Variable-length binary data (up to n bytes)                          | file_data VARBINARY(500)               |
| ENUM         | Predefined list of values, stores one of the possible options        | gender ENUM('Male', 'Female', 'Other') |
| SET          | Predefined list of values, stores multiple options (comma-separated) | tags SET('Science', 'Math', 'History') |
| JSON         | Stores JSON (JavaScript Object Notation) formatted data              | preferences JSON                       |

# CREATE - Define New Structures

Used to create tables, views, or other objects in the database.

```
CREATE DATABASE School;
```

```
CREATE TABLE table_name (  
    column1 datatype constraint,  
    column2 datatype constraint,  
    ... );
```

```
CREATE VIEW ActiveStudents AS  
SELECT StudentID, FirstName, LastName  
FROM Students  
WHERE EnrollmentDate IS NOT NULL;
```

# ALTER - Modify Existing Structures

Used to add, modify, or remove columns and constraints.

```
ALTER TABLE table_name  
ADD column_name datatype;
```

```
ALTER TABLE table_name  
DROP COLUMN column_name;
```

```
ALTER TABLE table_name  
RENAME COLUMN old_name to new_name;
```

```
ALTER TABLE table_name  
MODIFY COLUMN column_name datatype;
```



# DROP, TRUNCATE, RENAME

Used to delete database objects like tables, views, or databases permanently.

```
DROP TABLE table_name;
```

```
TRUNCATE TABLE table_name;
```

```
RENAME TABLE old_name TO new_name ;
```